

Topic 3: Principal Component Analysis*Lecturer: Dr. Kai Liu**Scribe: Claude*

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Principal component analysis (PCA) summarizes high-dimensional data with a few directions. We set it up as a least-squares reconstruction problem, show that it is equivalent to maximizing projected variance, and solve it with an eigendecomposition (or, equivalently, an SVD). We then look at what PCA finds in images of handwritten digits and close with sparse PCA. The prerequisites are the trace, the SVD, and positive semidefinite (PSD) matrices from the earlier linear algebra lecture. All vectors and matrices are real.

3.1 Notation and the goal

We observe n data points $x_1, \dots, x_n \in \mathbb{R}^d$, one per sample. In this lecture each data point is a *column*, so the data matrix is

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \in \mathbb{R}^{d \times n}.$$

Many software libraries store samples as *rows* instead ($n \times d$). Every formula below then holds with X replaced by its transpose, so it is worth checking which convention your code uses.

Symbol	Meaning	Size
n, d, r	number of samples, features, and components ($r \leq d$)	scalars
x_i	the i th data point	$d \times 1$
$\bar{x} = \frac{1}{n} \sum_i x_i$	the mean data point	$d \times 1$
$\tilde{x}_i = x_i - \bar{x}$	the centered data point	$d \times 1$
$\tilde{X} = [\tilde{x}_1 \cdots \tilde{x}_n]$	the centered data matrix	$d \times n$
$W = [w_1 \cdots w_r]$	orthonormal directions, $W^\top W = I_r$	$d \times r$
z_i	the code (coordinates) of x_i along W	$r \times 1$
$S = \tilde{X} \tilde{X}^\top = \sum_i \tilde{x}_i \tilde{x}_i^\top$	the scatter matrix	$d \times d$
$\lambda_1 \geq \cdots \geq \lambda_d \geq 0$	eigenvalues of S , eigenvectors $u_1, \dots, u_d$	scalars

Two renamings from the board. The handwritten notes call the codes λ_i ; we write z_i so that λ can denote eigenvalues. The notes also write the eigenvalues of $XX^\top$ as σ_i ; since we use σ_i for singular values of the data matrix, here they are λ_i . Dividing S by n gives the (population) covariance matrix; the constant does not change any of the optimal directions below.

3.1.1 Coordinates in an orthonormal basis

Every vector is a combination of basis vectors. In the standard basis of $\mathbb{R}^3$,

$$\begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} = 3 \underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}_{v_1} + 5 \underbrace{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}}_{v_2} + 4 \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{v_3}, \quad V = [v_1 \ v_2 \ v_3], \quad V^\top V = I.$$

When the columns of V are orthonormal, the coordinates of a vector x are simply $V^\top x$: the j th coordinate is the inner product $v_j^\top x$. No linear system has to be solved. This is why PCA insists on orthonormal directions.

Goal of PCA

Find $r \ll d$ orthonormal directions $W = [w_1 \cdots w_r]$, $W^\top W = I_r$, such that every data point is well approximated using only r coordinates along these directions. Example: a 100×100 grayscale image is a vector with $d = 10,000$ entries, yet a few dozen numbers often describe it well.

3.2 PCA as a reconstruction problem

Model each data point as an offset plus a combination of the r directions:

$$x_i \approx Wz_i + b, \quad W \in \mathbb{R}^{d \times r}, \quad W^\top W = I_r, \quad z_i \in \mathbb{R}^r, \quad b \in \mathbb{R}^d. \quad (3.1)$$

The vector b is an offset shared by all points, and z_i is the code of the i th point. Choose everything to minimize the total squared reconstruction error:

$$\min_{W^\top W = I_r, b, z_1, \dots, z_n} \sum_{i=1}^n \|x_i - (Wz_i + b)\|_2^2. \quad (3.2)$$

3.2.1 Why we need the offset, and why it is the mean

The directions in W pass through the origin, but data clouds usually do not. The offset moves the origin to the middle of the data. It is not unique, as the board example shows:

$$\begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} = 3e_1 + 4e_2 + 5e_3 + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 2e_1 + 3e_2 + 4e_3 + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Shifting the offset changes the coordinates but not the point. Shifting also leaves the spread unchanged: the numbers 3, 4, 5 and 6, 7, 8 both have variance $\frac{1}{3}((-1)^2 + 0^2 + 1^2) = \frac{2}{3}$. PCA only cares about the spread around the center, which suggests choosing $b = \bar{x}$.

Proposition 3.1 *For any fixed W with $W^\top W = I_r$, the offset $b = \bar{x}$ is optimal in (3.2).*

Proof. Write $P = WW^\top$. For fixed W and b , the best code is $z_i = W^\top(x_i - b)$ (shown in Section 3.3), leaving the residual $(I - P)(x_i - b)$. Split $x_i - b = (x_i - \bar{x}) + (\bar{x} - b)$ and expand the square. The cross terms add up to $2(\bar{x} - b)^\top (I - P)^\top (I - P) \sum_i (x_i - \bar{x})$, which is zero because $\sum_i (x_i - \bar{x}) = 0$. Therefore

$$\sum_{i=1}^n \|(I - P)(x_i - b)\|_2^2 = \sum_{i=1}^n \|(I - P)\tilde{x}_i\|_2^2 + n \|(I - P)(\bar{x} - b)\|_2^2,$$

and the last term vanishes when $b = \bar{x}$. ■

The proof also shows exactly how b fails to be unique: adding to $\bar{x}$ any vector in the span of W gives the same error. The standard choice is $b = \bar{x}$.

Worked example (from the board). With $x_1 = (2, 3, 4)^\top$ and $x_2 = (4, 1, 6)^\top$, the mean is $b = \bar{x} = (3, 2, 5)^\top$. The centered points are $\tilde{x}_1 = (-1, 1, -1)^\top$ and $\tilde{x}_2 = (1, -1, 1)^\top$. We finish this example in Section 3.9.

From now on we center first, $\tilde{x}_i = x_i - \bar{x}$, and solve

$$\min_{W^T W = I_r, z_1, \dots, z_n} \sum_{i=1}^n \|\tilde{x}_i - W z_i\|_2^2. \quad (3.3)$$

3.3 The best codes are projections

Fix W . Each term of (3.3) is a separate least-squares problem in z_i with design matrix W . Its normal equations read $W^T W z_i = W^T \tilde{x}_i$, and since $W^T W = I_r$,

$$\boxed{z_i = W^T \tilde{x}_i}, \quad W z_i = W W^T \tilde{x}_i. \quad (3.4)$$

The code collects the inner products with the directions, exactly as in the basis example. The matrix $P = W W^T$ is the orthogonal projection onto the span of W : it is symmetric and $P^2 = W(W^T W)W^T = P$. Note that $W W^T \neq I_d$ when $r < d$, even though $W^T W = I_r$.

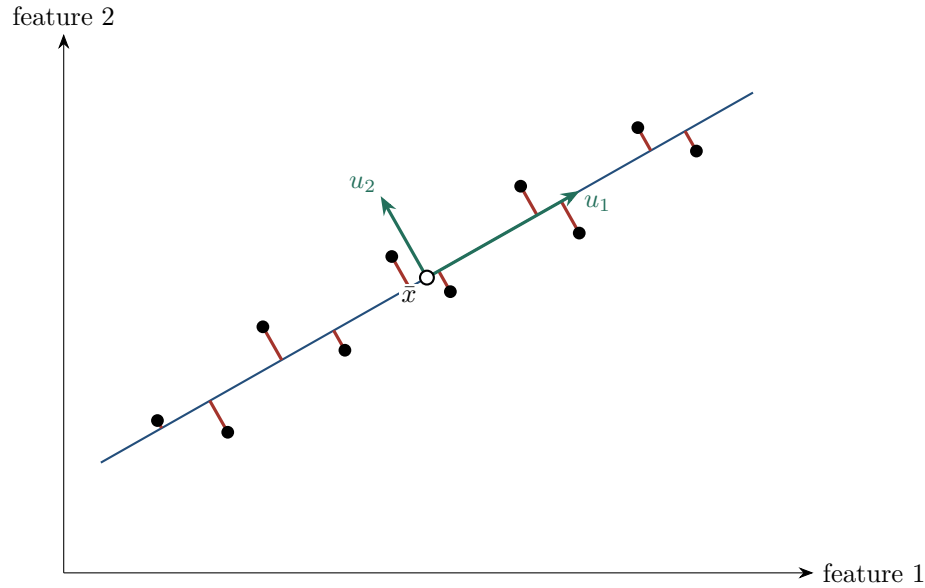


Figure 3.1. Ten points in two dimensions (black dots). The blue line passes through the mean $\bar{x}$ (open circle) along the first principal direction u_1 ; the short red segments are the residuals $\tilde{x}_i - u_1 u_1^T \tilde{x}_i$, which are perpendicular to the line. PCA picks the line that makes the sum of squared red lengths as small as possible. Here u_1 captures about 98% of the variance, and the second direction u_2 is perpendicular to it.

Unlike the least-squares regression from the earlier lecture, the residuals here are perpendicular to the line, not vertical: PCA treats all features alike and does not single out a response variable.

3.4 From reconstruction error to projected variance

Substituting the optimal codes (3.4) into (3.3) leaves a problem in W alone:

$$\min_{W^T W = I_r} \sum_{i=1}^n \|\tilde{x}_i - W W^T \tilde{x}_i\|_2^2. \quad (3.5)$$

Expand one term, writing x for $\tilde{x}_i$ to lighten notation:

$$\begin{aligned}\|x - WW^\top x\|_2^2 &= (x - WW^\top x)^\top (x - WW^\top x) \\ &= x^\top x - x^\top WW^\top x - x^\top WW^\top x + x^\top W \underbrace{(W^\top W)}_{=I_r} W^\top x \\ &= x^\top x - x^\top WW^\top x = \|x\|_2^2 - \|W^\top x\|_2^2.\end{aligned}$$

The constraint $W^\top W = I_r$ is what makes the last two terms combine. The result is the Pythagorean identity: the squared length of x splits into the part kept by the projection and the part lost.

The first term, $\sum_i \|\tilde{x}_i\|_2^2$, does not depend on W . Minimizing the error is therefore the same as maximizing what is kept:

$$\min_{W^\top W = I_r} \sum_{i=1}^n \|\tilde{x}_i - WW^\top \tilde{x}_i\|_2^2 \iff \max_{W^\top W = I_r} \sum_{i=1}^n \tilde{x}_i^\top WW^\top \tilde{x}_i = \max_{W^\top W = I_r} \sum_{i=1}^n \|z_i\|_2^2. \quad (3.6)$$

Since the codes $z_i = W^\top \tilde{x}_i$ have mean zero, $\frac{1}{n} \sum_i \|z_i\|_2^2$ is the total variance of the data after projecting onto W .

Two views of the same problem

Minimizing the squared reconstruction error is equivalent to maximizing the variance of the projected data. Good directions are the ones along which the data spread out the most.

3.5 Rewriting the objective with the trace

Each term $\tilde{x}_i^\top WW^\top \tilde{x}_i$ is a scalar, and a scalar equals its own trace. Using the cyclic property $\text{tr}(AB) = \text{tr}(BA)$ with $A = \tilde{x}_i^\top W$ ($1 \times r$) and $B = W^\top \tilde{x}_i$ ($r \times 1$),

$$\sum_{i=1}^n \tilde{x}_i^\top WW^\top \tilde{x}_i = \sum_{i=1}^n \text{tr}(W^\top \tilde{x}_i \tilde{x}_i^\top W) = \text{tr}\left(W^\top \left[\sum_{i=1}^n \tilde{x}_i \tilde{x}_i^\top\right] W\right) = \text{tr}(W^\top \tilde{X} \tilde{X}^\top W).$$

The middle step uses linearity of the trace, and the last step uses the outer-product view of a matrix product, $\tilde{X} \tilde{X}^\top = \sum_i \tilde{x}_i \tilde{x}_i^\top$. PCA is therefore

$$\max_{W \in \mathbb{R}^{d \times r}, W^\top W = I_r} \text{tr}(W^\top S W), \quad S = \tilde{X} \tilde{X}^\top. \quad (3.7)$$

The constraint is essential. Without $W^\top W = I_r$, replacing W by cW multiplies the objective by c^2 , so the maximum would be infinite. The constraint fixes the length of each direction and keeps the directions from repeating one another.

Only the subspace matters. If Q is an $r \times r$ orthogonal matrix, then WQ is also feasible and $(WQ)(WQ)^\top = WW^\top$, so the projection, the reconstruction error, and the objective are unchanged. PCA determines the best r -dimensional subspace; the particular basis is a convention.

A nonconvex problem with an exact answer. The feasible set $\{W : W^\top W = I_r\}$ is curved, not convex, so standard convex optimization does not apply directly. Nevertheless, the next two sections show that the global maximizer can be written down exactly from the eigenvectors of S .

3.6 Facts about symmetric matrices

The scatter matrix $S = \tilde{X}\tilde{X}^\top$ is symmetric. It is also PSD, because $v^\top S v = \|\tilde{X}^\top v\|_2^2 \geq 0$, so all its eigenvalues are nonnegative.

Lemma 3.2 *Let $A = A^\top$. If $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$ with $\lambda_1 \neq \lambda_2$, then $\langle v_1, v_2 \rangle = v_1^\top v_2 = 0$.*

Proof. Multiply the first equation on the left by $v_2^\top$ and the second by $v_1^\top$:

$$v_2^\top A v_1 = \lambda_1 v_2^\top v_1, \quad v_1^\top A v_2 = \lambda_2 v_1^\top v_2.$$

The left sides are equal: $v_2^\top A v_1$ is a scalar, so it equals its transpose $v_1^\top A^\top v_2 = v_1^\top A v_2$ by symmetry. Subtracting gives $(\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle = 0$, and since $\lambda_1 \neq \lambda_2$, we get $\langle v_1, v_2 \rangle = 0$. ■

Symmetry is used exactly once, in $A^\top = A$; for a general square matrix the conclusion can fail.

The full statement is the **spectral theorem**: every real symmetric S can be written

$$S = U \Lambda U^\top = \sum_{j=1}^d \lambda_j u_j u_j^\top, \quad U^\top U = U U^\top = I_d, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d), \quad (3.8)$$

with an orthonormal basis of eigenvectors $u_1, \dots, u_d$. When eigenvalues repeat, the lemma says nothing about vectors within one eigenspace, but an orthonormal basis of that eigenspace can still be chosen. Because S is PSD, (3.8) is also an SVD of S , which is how the board writes it: $XX^\top = U \Sigma U^\top$.

3.7 Solving PCA

3.7.1 One direction

For $r = 1$, problem (3.7) reads $\max_{\|v\|_2=1} v^\top S v$. Expand v in the eigenvector basis, $v = \sum_j c_j u_j$; then $\|v\|_2^2 = \sum_j c_j^2 = 1$ and, by (3.8),

$$v^\top S v = \sum_{j=1}^d \lambda_j c_j^2 \leq \lambda_1 \sum_{j=1}^d c_j^2 = \lambda_1. \quad (3.9)$$

The objective is a weighted average of the eigenvalues with weights c_j^2 summing to one, so it can never exceed the largest one.

Taking $v = u_1$ attains the bound. Using orthonormality, $u_1^\top u_1 = 1$ and $u_1^\top u_j = 0$ for $j \neq 1$:

$$u_1^\top S u_1 = u_1^\top \left(\sum_{j=1}^d \lambda_j u_j u_j^\top \right) u_1 = \sum_{j=1}^d \lambda_j (u_1^\top u_j) (u_j^\top u_1) = \lambda_1.$$

A mixture does worse. The unit vector $v = 0.8u_1 + 0.6u_2$ (note $0.8^2 + 0.6^2 = 1$) gives $v^\top S v = 0.64\lambda_1 + 0.36\lambda_2$, which is smaller than λ_1 whenever $\lambda_2 < \lambda_1$. Mixing in a weaker direction always costs variance. So $v^* = u_1$: the first principal direction is the top eigenvector of the scatter matrix.

3.7.2 Several directions

Theorem 3.3 Let $S = \sum_j \lambda_j u_j u_j^\top$ with $\lambda_1 \geq \dots \geq \lambda_d$. For every $W \in \mathbb{R}^{d \times r}$ with $W^\top W = I_r$,

$$\text{tr}(W^\top S W) \leq \lambda_1 + \dots + \lambda_r,$$

with equality for $W = U_r = [u_1 \dots u_r]$, the first r columns of U (written $U(:, 1:r)$ in MATLAB notation).

Proof. Set $h_j = \|W^\top u_j\|_2^2 \geq 0$. By (3.8) and the trace identity of Section 3.5,

$$\text{tr}(W^\top S W) = \sum_{j=1}^d \lambda_j \text{tr}(W^\top u_j u_j^\top W) = \sum_{j=1}^d \lambda_j h_j.$$

The weights obey two constraints. First, $h_j = u_j^\top (W W^\top) u_j \leq \|u_j\|_2^2 = 1$, because $W W^\top$ is a projection and a projection never lengthens a vector. Second, $\sum_j h_j = \text{tr}(W^\top U U^\top W) = \text{tr}(W^\top W) = r$. To maximize $\sum_j \lambda_j h_j$ subject to $0 \leq h_j \leq 1$ and $\sum_j h_j = r$, put weight one on the r largest eigenvalues; any other allocation moves weight to a smaller eigenvalue. For $W = U_r$, one checks $h_j = 1$ for $j \leq r$ and $h_j = 0$ otherwise, so the bound is attained. ■

The PCA solution

The optimal directions are the top r eigenvectors of $S = \tilde{X} \tilde{X}^\top$: $W^* = U_r = [u_1 \dots u_r]$. The codes are $z_i = U_r^\top \tilde{x}_i$, and the reconstruction is $\hat{x}_i = \bar{x} + U_r U_r^\top \tilde{x}_i$.

Combining Section 3.4 with the theorem, and using $\sum_i \|\tilde{x}_i\|_2^2 = \text{tr}(S) = \sum_j \lambda_j$, the smallest achievable error is

$$\min_{W^\top W = I_r} \sum_{i=1}^n \|\tilde{x}_i - W W^\top \tilde{x}_i\|_2^2 = \sum_{j=r+1}^d \lambda_j. \quad (3.10)$$

The **fraction of variance explained** by r components is $(\lambda_1 + \dots + \lambda_r) / (\lambda_1 + \dots + \lambda_d)$; plotting it against r is the usual way to choose r (Figure 3.3).

3.8 Computing PCA in practice

3.8.1 Through the SVD of the data

Let $\tilde{X} = U \Sigma V^\top$ be an SVD with singular values $\sigma_1 \geq \sigma_2 \geq \dots$. Then

$$S = \tilde{X} \tilde{X}^\top = U \Sigma V^\top V \Sigma^\top U^\top = U (\Sigma \Sigma^\top) U^\top,$$

so the eigenvectors of S are the left singular vectors of $\tilde{X}$ and $\lambda_j = \sigma_j^2$. The codes are also available directly: $Z = U_r^\top \tilde{X} = \Sigma_r V_r^\top$. Software usually computes PCA this way, because forming S squares the singular values and can lose accuracy.

Step	PCA with r components
1	Compute the mean $\bar{x}$ and center: $\tilde{x}_i = x_i - \bar{x}$.
2	Compute the top r left singular vectors U_r of $\tilde{X}$ (equivalently, the top r eigenvectors of S).
3	Codes: $z_i = U_r^\top \tilde{x}_i$. Reconstructions: $\hat{x}_i = \bar{x} + U_r z_i$.
4	Report the explained variance $\sum_{j \leq r} \sigma_j^2 / \sum_j \sigma_j^2$.

A new point x is encoded with the *training* mean: $z = U_r^\top (x - \bar{x})$.

3.8.2 Images and very large d

A 100×100 grayscale image becomes a vector of length $d = 10,000$ by stacking its pixels. The scatter matrix is then $10,000 \times 10,000$, with 10^8 entries. When there are fewer images than pixels ($n < d$), the thin SVD of the $d \times n$ matrix $\tilde{X}$ avoids forming S at all, and at most $n - 1$ eigenvalues are nonzero after centering.

3.8.3 Things that often go wrong

- **Forgetting to center.** Without centering, the top direction often just points at the mean rather than along the spread.
- **Features on different scales.** PCA is driven by variance, so a feature measured in millimeters dominates the same feature measured in meters. Standardize features when their units are not comparable.
- **Signs are arbitrary.** If u_j is a principal direction, so is $-u_j$; different software may return either. When eigenvalues tie, any orthonormal basis of the tied eigenspace is equally valid.
- **Variance is not the same as usefulness.** PCA ignores labels, so the directions of largest spread need not separate classes.

3.9 Examples

3.9.1 Finishing the board example

For $\tilde{x}_1 = (-1, 1, -1)^\top$ and $\tilde{x}_2 = (1, -1, 1)^\top$,

$$S = \tilde{x}_1 \tilde{x}_1^\top + \tilde{x}_2 \tilde{x}_2^\top = 2 \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} = 6 u_1 u_1^\top, \quad u_1 = \frac{1}{\sqrt{3}}(1, -1, 1)^\top.$$

So $\lambda_1 = 6$ and $\lambda_2 = \lambda_3 = 0$. With $r = 1$, the codes are $z_1 = u_1^\top \tilde{x}_1 = -\sqrt{3}$ and $z_2 = \sqrt{3}$, and

$$\bar{x} + u_1 z_1 = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = x_1.$$

The reconstruction is exact, as (3.10) predicts: the lost variance $\lambda_2 + \lambda_3$ is zero, because two points always lie on one line.

3.9.2 Handwritten digits

The figures below use the MNIST training set: 60,000 images of handwritten digits, each 28×28 pixels with values in $[0, 1]$, so $d = 784$. PCA was computed on all 60,000 images. An interactive version of these figures, “What PCA sees in handwritten digits,” lets you move through the code space and rebuild digits from any number of components.

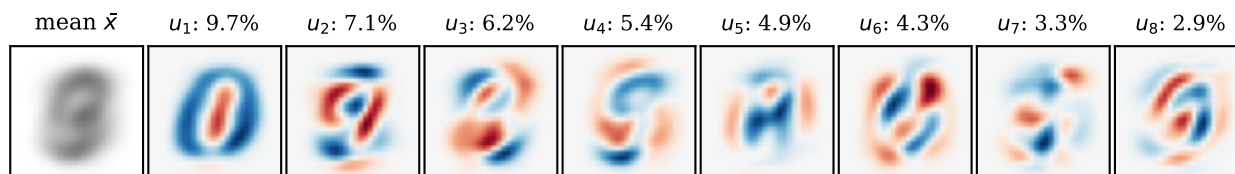


Figure 3.2. The mean digit $\bar{x}$ (left) and the first eight principal directions, each shown as a 28×28 image with the share of total variance it explains. Blue pixels have positive weight and red pixels negative weight: moving in the $+u_j$ direction adds ink where u_j is blue and removes it where u_j is red. The mean is a blur of all digits, and u_1 roughly contrasts a “0”-like ring with a central vertical stroke.

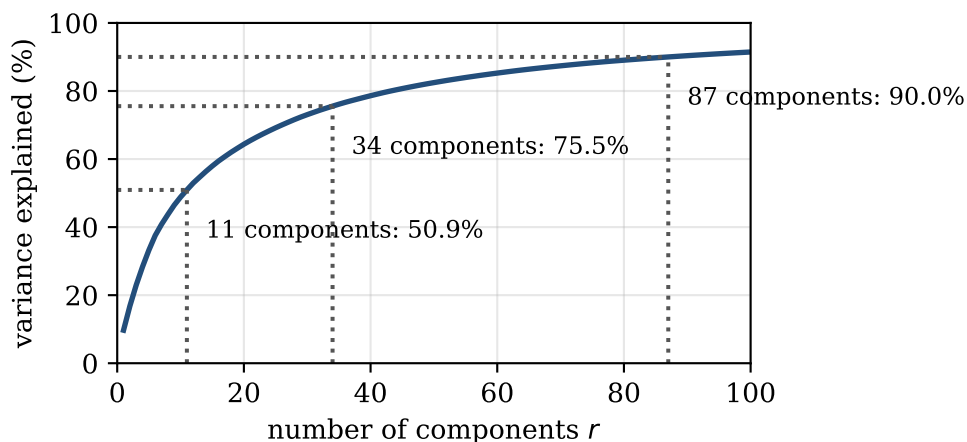


Figure 3.3. Fraction of the total pixel variance explained by the first r components. One component explains 9.7% and two explain 16.8%; reaching 50%, 75%, and 90% takes 11, 34, and 87 components out of 784.

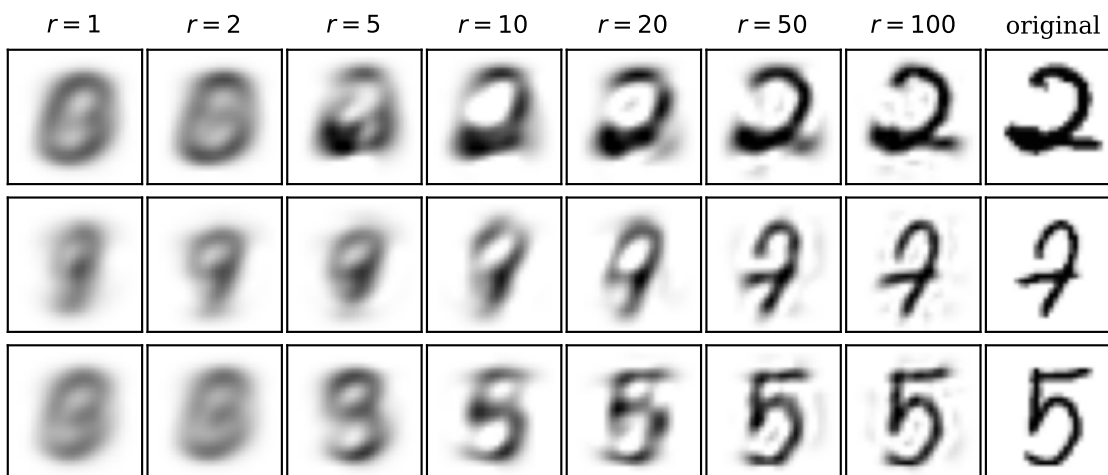


Figure 3.4. Reconstructions $\hat{x} = \bar{x} + U_r U_r^\top (x - \bar{x})$ of three training digits (rows) from $r = 1$ to $r = 100$ components, next to the original (right). With one or two components every digit looks like a blurred average; around $r = 20$ to 50 the digit becomes recognizable while using far fewer than 784 numbers.

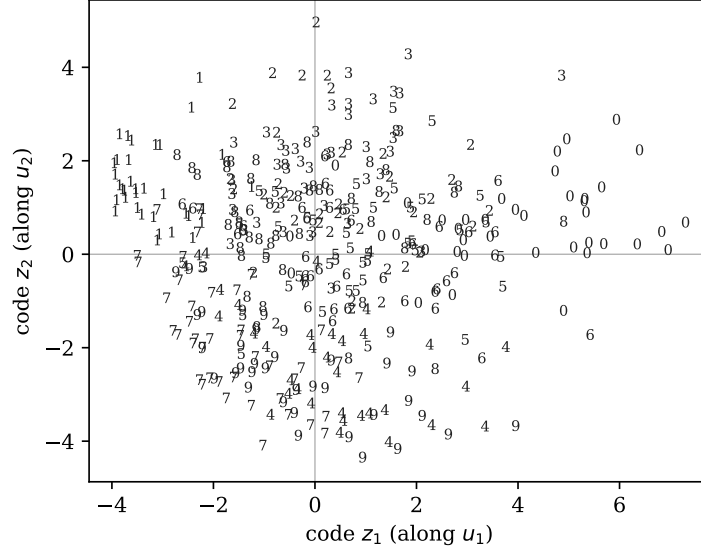


Figure 3.5. The first two codes (z_1, z_2) of 400 training images, each plotted as its digit label. Some digits occupy distinct regions (zeros on the right, ones on the left), while others overlap heavily. Two directions keep only 16.8% of the variance, and PCA never sees the labels, so it does not try to separate the classes.

3.10 Sparse PCA

Each principal direction is usually *dense*: it mixes all d features, which makes it hard to interpret. In a gene-expression study, for example, one would rather have each component involve a handful of genes. Sparse PCA adds an ℓ_1 penalty to the reconstruction problem:

$$\min_{W^\top W = I_r} \frac{1}{2} \left\| \tilde{X} - W W^\top \tilde{X} \right\|_F^2 + \gamma \|W\|_1, \quad \|W\|_1 = \sum_{j,k} |W_{jk}|. \quad (3.11)$$

The board writes the penalty weight as λ ; we use $\gamma \geq 0$ to avoid a clash with the eigenvalues. The first term is (3.5) written for the whole matrix, since $\left\| \tilde{X} - W W^\top \tilde{X} \right\|_F^2 = \sum_i \|\tilde{x}_i - W W^\top \tilde{x}_i\|_2^2$. As in Section 3.4, it equals $\frac{1}{2} \text{tr}(S) - \frac{1}{2} \text{tr}(W^\top S W)$.

Two extremes.

- With $\gamma = 0$ we recover ordinary PCA, solved by $W = U_r$.
- The sparsest possible orthonormal directions are standard basis vectors. For example, with $d = r = 3$,

$$W = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is orthonormal, and each “component” is simply one original feature.

To see why the second case is the extreme, note that a unit vector w satisfies $\|w\|_1 \geq \|w\|_2 = 1$, with equality exactly when w has a single nonzero entry equal to ± 1 . Hence $\|W\|_1 \geq r$ for every feasible W , and the minimum is attained by signed standard basis vectors. Large γ pushes the solution toward such directions, trading explained variance for interpretability.

Unlike PCA, problem (3.11) has no closed-form solution in general when $\gamma > 0$; it is solved with iterative methods that are beyond this lecture.

3.11 Summary and review questions

Object	Formula or fact
Model	$x_i \approx Wz_i + b$ with $W^\top W = I_r$.
Offset	$b = \bar{x}$ is optimal; center the data first.
Codes	$z_i = W^\top \tilde{x}_i$; reconstruction $\bar{x} + WW^\top \tilde{x}_i$.
Equivalence	$\min \sum_i \ \tilde{x}_i - WW^\top \tilde{x}_i\ _2^2 \Leftrightarrow \max \text{tr}(W^\top SW)$, $S = \tilde{X}\tilde{X}^\top$.
Solution	$W^* = U_r$, the top r eigenvectors of S (left singular vectors of $\tilde{X}$).
Optimal values	kept variance $\sum_{j \leq r} \lambda_j$; lost variance $\sum_{j > r} \lambda_j$; $\lambda_j = \sigma_j^2$.
Symmetric matrices	eigenvectors for distinct eigenvalues are orthogonal.
Sparse PCA	add $\gamma \ W\ _1$; $\gamma = 0$ gives PCA, no closed form for $\gamma > 0$.

3.11.1 Review questions

1. Show that $W^\top W = I_r$ does not imply $WW^\top = I_d$ when $r < d$. What is $WW^\top$ geometrically?
2. In the expansion of $\|x - WW^\top x\|_2^2$, which step fails if the columns of W are not orthonormal?
3. Compute the variance of 3, 4, 5 and of 6, 7, 8. Why does PCA subtract the mean before looking for directions?
4. For the points $x_1 = (2, 3, 4)^\top$ and $x_2 = (4, 1, 6)^\top$, verify that $\bar{x} + u_1 z_2$ reproduces x_2 .
5. Let $S = \text{diag}(5, 2, 1)$. Find the best direction v and its objective value. Evaluate $v^\top S v$ for $v = (0.8, 0.6, 0)^\top$ and compare.
6. For $r = 2$ and the same S , give the optimal W , the kept variance, and the reconstruction error. What fraction of variance is explained?
7. Explain why the problem $\max \text{tr}(W^\top S W)$ has no finite solution if the constraint $W^\top W = I_r$ is removed.
8. A dataset has $n = 500$ images of size 100×100 . What are the sizes of $\tilde{X}$ and S ? At most how many eigenvalues of S are nonzero?
9. Why is $\|W\|_1 \geq r$ for every W with orthonormal columns? Describe all W achieving equality.